Week 6 - Reading Week Consolidation: Worked Answers

EART23001 Atmospheric Physics and Weather

Workload guide. The shaded question headings show the intended workload:

- **IN CLASS**: selected anchor problems that we may work through together in the timetabled session.
- **YOU SHOULD TRY**: expected independent practice after the session.
- **EXTRA CHALLENGE**: optional problems for students who want additional depth or a harder application.

The categories are repeated here so the worked answers map directly onto the practice sheet. Use the solutions to check method, units and physical interpretation rather than only the final number. A course-wide Symbols, Constants and Units sheet is provided separately; non-standard symbols are also defined locally.

This is an optional consolidation sheet for Reading Week. It contains no new material and deliberately mixes ideas from Weeks 1–5.

YOU SHOULD TRY Question 1: A launch-day atmospheric state

At the surface, pressure is 1005 hPa and temperature is 285 K.

1. Calculate the air density using $R_d = 287\ \mathrm{J\,kg^{-1}\,K^{-1}}$.
2. Using a scale height of 8.2 km, estimate the pressure at 2.0 km.
3. Give one reason why the actual pressure at 2 km may differ from this simple estimate.

Worked answer

The surface density is

$$\rho = \frac{100500}{287 \times 285} = 1.23\ \mathrm{kg\,m^{-3}}.$$

The pressure at 2 km is

$$p = 1005 \exp(-2/8.2) = 787\ \mathrm{hPa}.$$

The exponential model assumes a constant scale height. In reality, temperature varies with height and time, and therefore the layer-mean scale height changes. Weather systems also alter the pressure field horizontally.

EXTRA CHALLENGE Question 2: Pressure as atmospheric mass

What fraction of the atmospheric mass lies above the 700 hPa pressure surface if mean sea-level pressure is 1013 hPa? What fraction lies below it?

Worked answer

Under hydrostatic balance, mass above a pressure level is proportional to pressure. Therefore

$$f_{above} = \frac{700}{1013} = 0.691.$$

About 69.1% of the atmospheric mass lies above the 700 hPa surface and

$$f_{below} = 1 - 0.691 = 0.309,$$

so about 30.9% lies below it.

YOU SHOULD TRY Question 3: From a pressure difference to a wind estimate

Across a 500 km distance, pressure decreases by 12 hPa. Take density $\rho = 1.10\ \mathrm{kg\,m^{-3}}$ and latitude 50°N.

1. Calculate the horizontal acceleration due to the *pressure-gradient force alone*,
$$|a_p| = \frac{1}{\rho}\left|\frac{\Delta p}{\Delta x}\right|.$$
This is not the parcel's total horizontal acceleration: once the air is moving, Coriolis and other forces may also contribute.
2. Calculate $f = 2\Omega \sin\phi$ using $\Omega = 7.292 \times 10^{-5}\ \mathrm{s^{-1}}$.
3. Assuming straight isobars and negligible friction, estimate the geostrophic wind speed by balancing the pressure-gradient and Coriolis accelerations.

Worked answer

$$|a_p| = \frac{1200}{1.10 \times 5.00 \times 10^5} = 2.18 \times 10^{-3}\ \mathrm{m\,s^{-2}}.$$

At 50°N,

$$f = 2(7.292 \times 10^{-5}) \sin 50^\circ = 1.12 \times 10^{-4}\ \mathrm{s^{-1}}.$$

Hence

$$v_g = \frac{|a_p|}{f} = \frac{2.18 \times 10^{-3}}{1.12 \times 10^{-4}} = 19.5\ \mathrm{m\,s^{-1}}.$$

This is an idealised upper-air estimate; friction and curvature alter real winds.

YOU SHOULD TRY Question 4: Put the dynamical chain in the correct order

A sunny coastline develops a sea breeze. The following statements are deliberately out of order:

- a near-surface pressure-gradient acceleration develops from sea toward land;
- the land warms faster than the sea;
- a closed circulation develops, with a return flow aloft;
- the air column over land becomes warmer and its pressure decreases more slowly with height;
- near-surface air begins to move onshore.

Put the statements into a physically sensible order. Then state whether Coriolis deflection would be more important for a weak local sea breeze lasting a few hours or for a synoptic-scale pressure system lasting several days, and explain why.

Worked answer

A sensible sequence is:

1. the land warms faster than the sea;
2. the air column over land becomes warmer and its pressure decreases more slowly with height;
3. a near-surface pressure-gradient acceleration develops from sea toward land;
4. near-surface air begins to move onshore;
5. a closed circulation develops, with a return flow aloft.

Coriolis deflection is generally more important for the larger, longer-lived synoptic-scale flow because the air travels for longer times and distances while the Earth rotates beneath it. A small, short-lived sea-breeze circulation can be strongly affected by local pressure gradients, friction and coastline geometry before Coriolis has time to produce a near-geostrophic balance.

EXTRA CHALLENGE Question 5: Which approximation is being used?

For each statement, identify the approximation or physical law being invoked, and give one circumstance in which it could fail or need refinement.

1. $p = \rho R_d T$.
2. $dp/dz = -\rho g$.
3. $p = p_0 e^{-z/H}$ with constant H.
4. Upper-level wind is parallel to straight isobars.

Worked answer

1. Ideal-gas equation of state for dry air. Moisture requires treating water vapour explicitly, and at very unusual pressures/temperatures real-gas effects may matter.
2. Hydrostatic balance. Strong convective updraughts, explosions or other rapidly accelerating vertical motions can depart from hydrostatic balance locally.
3. Constant-scale-height exponential atmosphere. It assumes a simplified temperature structure; real $T(z)$ makes H vary.
4. Geostrophic approximation. It neglects friction and curvature/time dependence; it is poor near the surface, near the equator, and in strongly accelerating/curved flows.

YOU SHOULD TRY Question 6: Exam-style synthesis

A weather map shows tightly packed isobars over northern Britain and widely spaced isobars over southern Britain. A radiosonde launched in the north shows strong winds at 2 km.

Explain, using the pressure-gradient and Coriolis forces, why strong winds are expected in the north. Then explain why the near-surface wind direction and speed need not match the 2 km wind exactly.

Worked answer

A weather map shows *tightly packed isobars over northern Britain and widely spaced isobars over southern Britain.* Tightly packed isobars mean a large horizontal pressure gradient $|\Delta p/\Delta x|$, so the pressure-gradient acceleration is large in the north. Away from strong friction, the wind accelerates until Coriolis acceleration becomes comparable, producing a relatively large geostrophic wind speed. The widely spaced isobars in the south imply a weaker pressure gradient and, other things being equal, weaker winds.

A radiosonde launched in the north shows *strong winds at* 2 km. This is consistent with the strong pressure gradient indicated by the tightly packed isobars.

The *near-surface wind direction and speed need not match the* 2 km *wind exactly.* Near the surface, friction slows the wind. Because Coriolis acceleration is proportional to wind speed, it becomes smaller while the pressure-gradient acceleration is essentially unchanged. The near-surface flow therefore crosses the isobars toward lower pressure and is usually slower than the wind above the frictionally influenced layer.